Community-Based Experience and Service (COBES)

A Case Report on Shao Community

NOBILIS PREAMBULUS (500 LEVEL MLS),
FACULTY OF BASIC CLINICAL SCIENCES,
UNIVERSITY OF ILORIN,
KWARA STATE,
NIGERIA.

Table of contents

01 Honourary Greetings

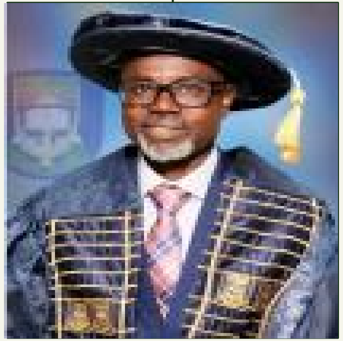
His Excellency
Abdulrasaq ABDULRAHMAN

Governor of Kwara State

President
Bola Ahmed TINUBU

President of Nigeria

Prof.
Wahab Olasupo EGBEWOLE



Vice Chancellor,
University of Ilorin

01 Honourary Greetings

Prof. B.S. ALABI

Provost, COHS

Prof. A.A. AKANBI II

Dean,
Faculty of Basic Clinical Sciences

Dr. G.O. ADUNMO

Head of Department,
Medical Laboratory Science

01 Honourary Greetings

Prof. A.O. Bolakale

COBES Coordinator

Oba Dr. Job Obalowu Atolagbe

Ohoro of Shao

Dr. T.D. Adeniyi

Site Supervisor

01 Honourary Greetings

Mr. Saadu Ibrahim Kolawale

Shao PHC Med. Lab. Scientist

Elder Job Bello of Onilude

COBES (Shao) Pathfinder

02

Group List

Group List (68)

- ABDULSALAM, Sherif Olalekan 19/46PW001
- ABDULWASII, Abdulbasit Opeyemi 19/46PW002
- ABE, Adedoyinsola Mercy 19/46PW003
- ABIOSE, Dhakiyah Mopelola 20/46PW091
- ADEBAYO, Rosemary Ayomide 20/46PW092
- ADEGOKE, Emmanuel Tolulope 20/46PW093
- ADENIJI, Islamiyat Eniola 19/46PW006
- ADESINA, Helen Anuoluwapo 19/46PW007
- ADETUNJI, Tesleem Oluwakayode 19/46PW008
- ADEWUYI, Aishat Adedamola 19/46PW009
- ADEYEMI, Tomilayo Lydia 19/46PW010
- ADISA, Islamiyah Temitope 20/46PW094
- ADUNSE, Christiana Oluwatimilehin 19/46PW011
- AFOLAYAN, Opeyemi Enoch 19/46PW012
- AKINOLA, Ibrahim Akinwale 19/46PW013
- ALASINRIN, Ruqayyah Olatinuke 19/46PW015
- ALAWONLA, Yetunde Arinola 19/46PW016
- ALBARIKA, Ololade Hiqmat 19/46PW017
- ALIYU, Thanni Oladimeji 19/46PW018
- AMINULLAHI, Islamiyat Ayodeji 20/46PW095
- AWONIYI, Oluwatimelehin Covenant 19/46PW020
- BAJO, Elizabeth Olayinka 20/46PW096
- BALOGUN, Rildwan Abiodun 19/46PW021
- BUSARI, Habeebat Olohunoyin 19/46PW022
- BUSARI, Kawthar Oyinkansola 19/46PW023
- DADA, Mathilda Kehinde 19/46PW024
- EKERE, Victoria Ihotu 19/46PW026
- ELUCHIE, Vivian Chimdindu 19/46PW027
- FAMUWAGUN, Toluwalope Serah 19/46PW028
- GIWA, Abdul-Hafiz Adedayo 19/46PW029
- HASSAN, Faridah Ajibola 19/46PW031
- IBITADE, Timothy Oluwaseun 19/46PW032
- IBRAHIM, Aishat Yetunde 19/46PW033
- IBRAHIM, Fuad Olamilekan 19/46PW034

Group List

- IJAIYA, Olayinka Sekinat 19/46PW035
- IMOSEMI, Eguavon Moses 19/46PW037
- ISIAKA, Busirat Iyabo 19/46PW038
- ISSA, Aishat Motunrayo 19/46PW039
- KADRI, Misturat Iyanu 19/46PW041
- LATEEF, Tunde Habeeb 19/46PW046
- LERAMOH, Omowunmi Victoria 19/46PW047
- MOSES, Deborah Oluwatomilola 19/46PW048
- OJEMEN, David Ebosetale 19/46PW053
- OJOAWO, Stephen Olusegun 19/46PW054
- OKUSANYA, Similoluwa Atinuke 19/46PW055
- OLAITAN, Hanifat Abake 20/46PW099
- OLAJIDE, Love Adeola 19/46PW057
- OLALERE, Nafisat Olabisi 20/46PW100
- OLAWUMI, Elizabeth Olajumoke 19/46PW058
- OLORUNFEMI, Mercy Oluwadamilola 19/46PW059
- OLUWAFEMI, Faith Daniel 19/46PW060
- OMIDIJI, Felicia Oluwabukunmi 19/46PW061
- OMOTAYO, Samuel Oluwadamilare 19/46PW062
- ONASANYA, Tomisin Sunday 19/46PW064
- OSENI, Basit Babatunde 17/25PB115
- OSHO, Mercy Adeola 19/46PW066
- OSUNDARA, Oluwadarasimi Razaq 19/46PW067
- OWOIYA, Toheeb Opeyemi 19/46PW068
- OYEDEMI, Oluwabemiwo Deborah 20/46PW103
- POPOOLA, Fatimoh Adebukola 19/46PW069
- RUNSEWE, Ayomide Oluwatobi 20/46PW104
- SAHEED, Habeeb Towobola 19/46PW071
- SALAMI, Faidat Oyinade 19/46PW072
- SALAMI, Kehinde Shakirat 20/46PW105
- SALAU, Favour Iteoluwa 19/46PW073
- SULAIMAN, Maryam Bolanle 19/46PW075
- SULEIMAN, Jamiu Babatunde 20/46PW106
- UBAH, Yagazie Gideon 19/46PW077

03

Introduction

Introduction to COBES & WHO

- The Community-Based Experience and Services (COBES) program is a student-centered initiative that immerses laboratory students in community health settings to understand local health practices, identify inadequacies, and implement solutions.
- Established in 1978 at the Faculty of Health Sciences, the program emphasizes problem-solving and collaboration within communities.

- The World Health Organization (WHO) marks its founding anniversary on April 7 with World Health Day, focusing on a major global health challenge each year.
- The 2025 theme, **"Healthy Beginnings, Hopeful Futures,"** aims to reduce preventable deaths, improve healthcare quality, support women's health, and mobilize communities for better public health.

Overview of Minimum Learning Objectives (MLOs)

MLO 1: Data on Health Facilities & Services

MLO 2: Challenges in Accessing Healthcare

MLO 3: Impediments from Health Service Providers & Facilities

MLO 4: Level of Government & Private Sector involvement

MLO 5: Sensitization & Health Promotion

MLO 6: Mobilizing Community Resources

MLO 7: Reporting & Presentation

MLS-Specific MLO: Access & Gaps in Medical Laboratory Services (MLS)

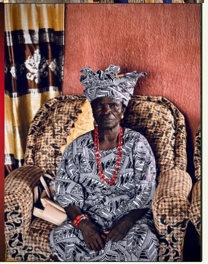
Arrival & Community Entry

- Visit to the King
- Visit to the Primary Healthcare Center (PHC)
- Visit to the wards of Shao Community

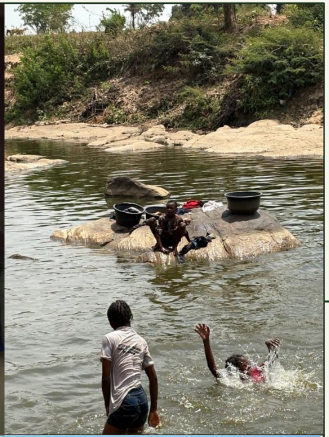
Arrival & Community Entry







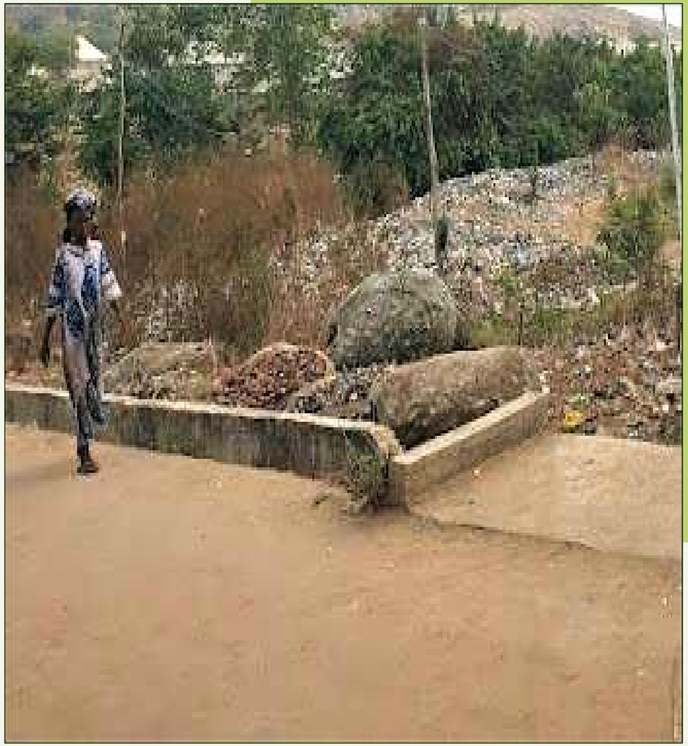
History of Shao


Demographic Survey

Methodology

- Each team ventured into various wards within the community to conduct surveys using Google Forms. A team member filled out the questionnaire strictly based on the responses provided by the responder.
- A total of 497 individuals participated, providing insights into age, gender, occupation, religion, ethnicity, household income, marital status, and education level. This section presents a comprehensive analysis of these factors.



Demographic Survey

Age Distribution

- Out of 497 respondents, the age structure of the Shao community skews toward younger adults, with 61% of the population falling below 40 years of age.

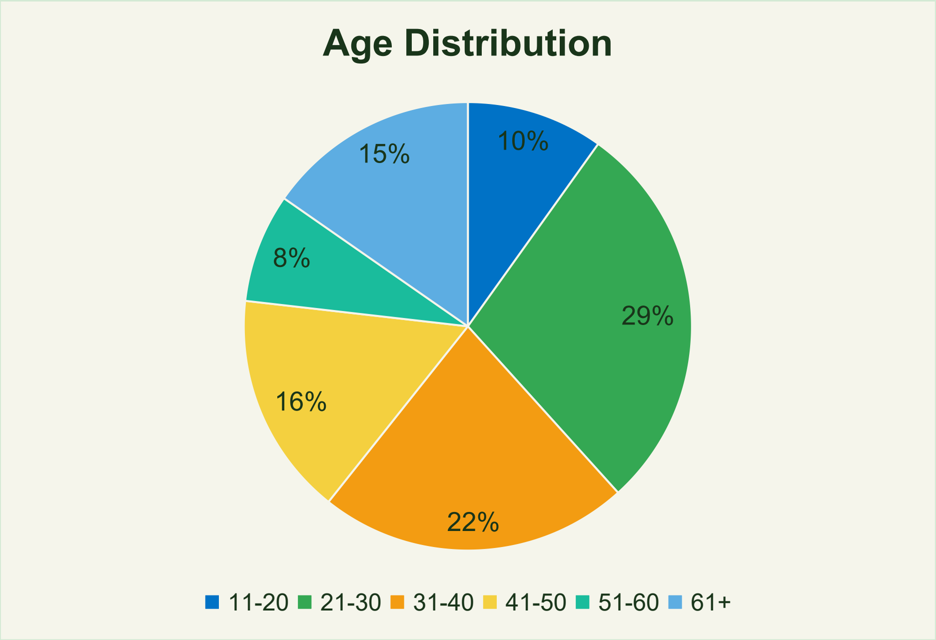


Figure 1: Pie Chart illustrating the age distribution in Shao

Demographic Survey

Sex Distribution

- In terms of sex distribution, the data reflect a relatively balanced but slightly female-majority population.

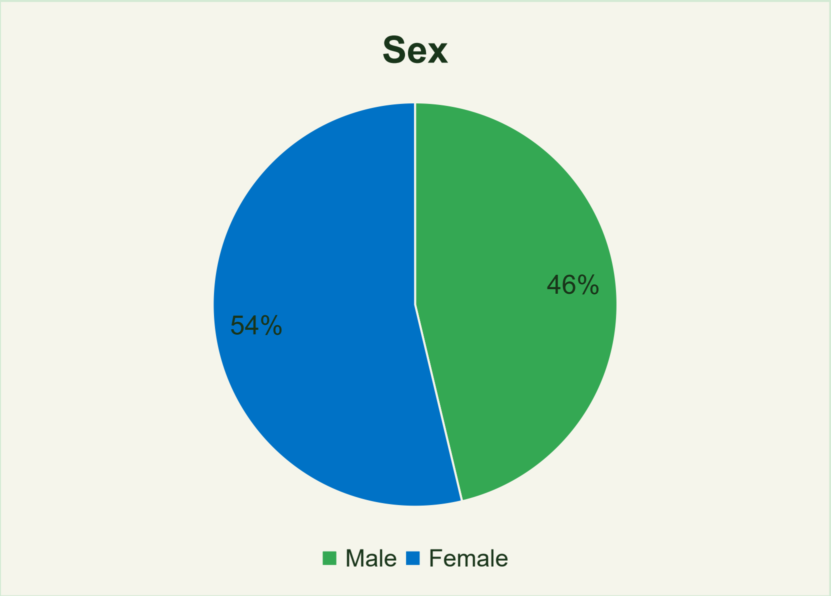


Figure 2: Pie Chart illustrating the sex distribution in Shao

Demographic Survey

Occupation and Income Levels

- Shao's economy is diverse, driven by farming, trading, civil service, skilled trades and other roles such as hunting, transport, and community health.
- Most households in Shao earn low-to-mid incomes, with 87% earning between ₦0 and ₦100,000 monthly. A small group earn above ₦100,000, often due to stable jobs or successful businesses.

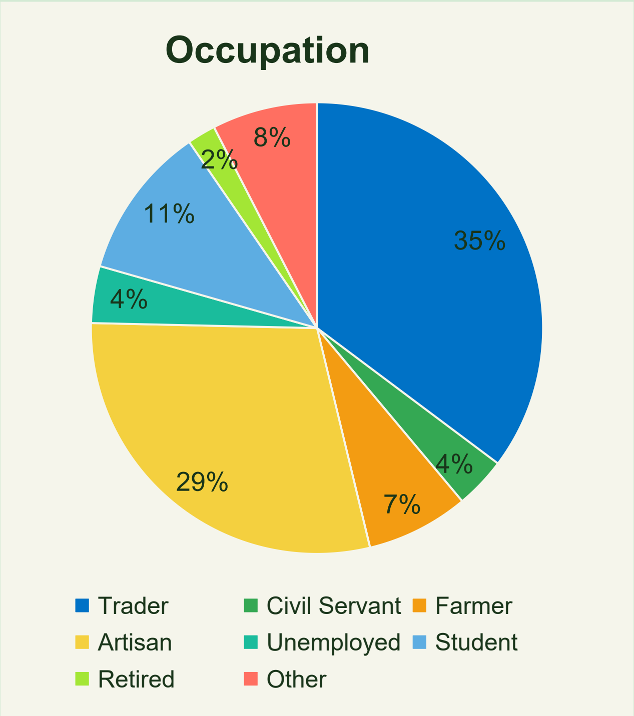


Figure 3: Pie Chart illustrating the diverse occupation in Shao

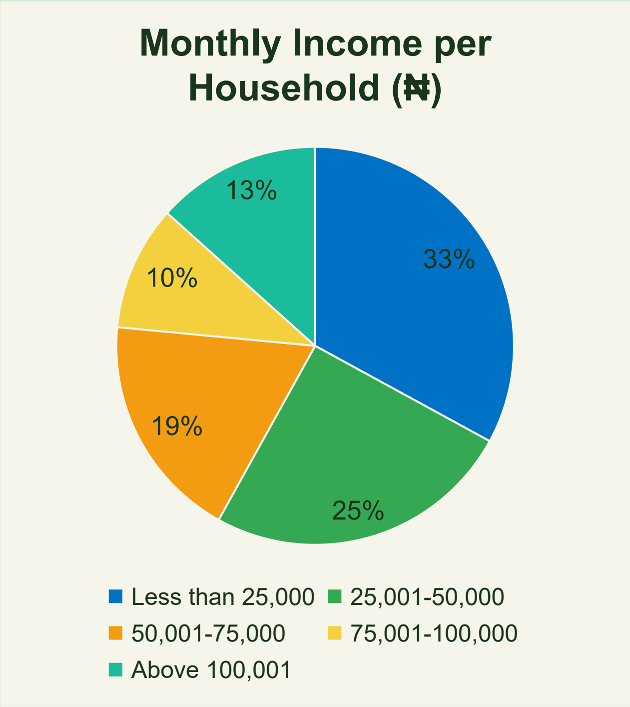


Figure 4: Pie Chart illustrating the income levels per household in Shao

Demographic Survey

Ethnicity

- The Shao community is largely composed of one dominant ethnic group—often Yoruba in many parts of Kwara State—alongside smaller groups such as Hausa, Igala, Tiv, Edo, Basa, and others.

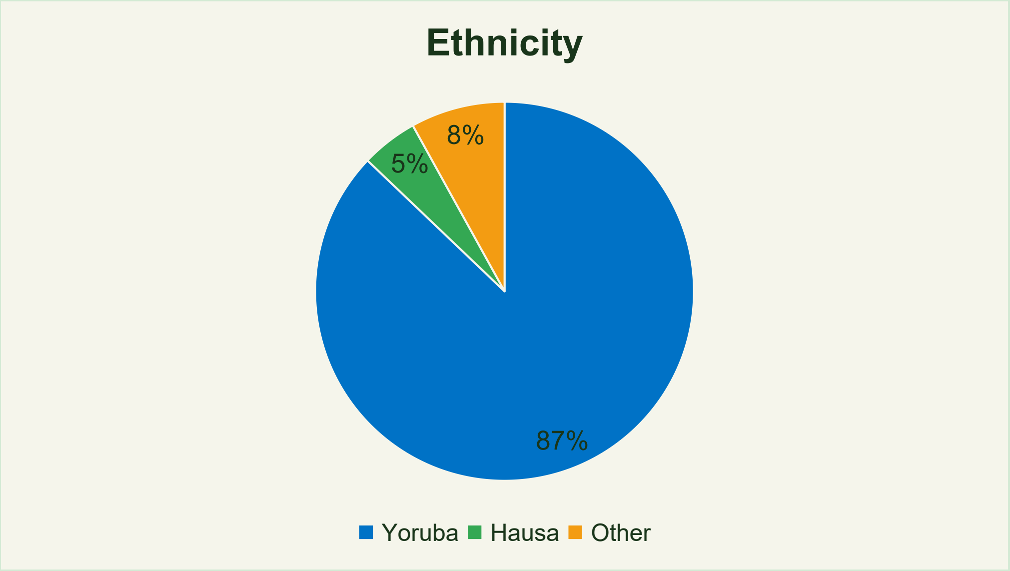


Figure 5: Pie Chart illustrating the ethnic groups in Shao

Demographic Survey

Religion

- Shao's religious landscape is diverse, with most following Christianity or Islam and a smaller group practicing Traditional religion.
- Interfaith harmony exists and this supports community development.

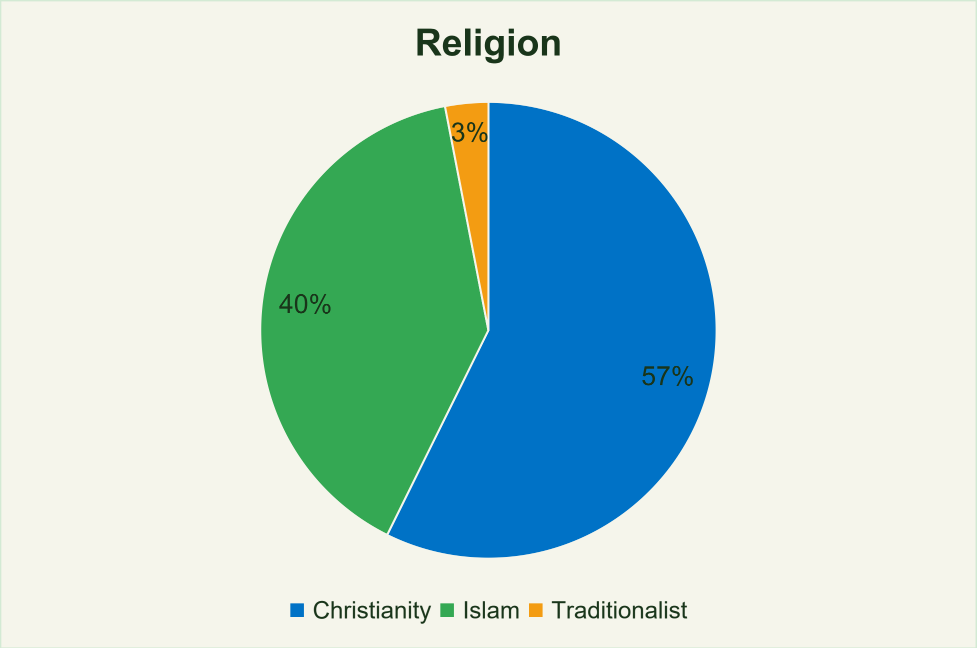


Figure 6: Pie Chart illustrating the religious landscape of Shao

Demographic Survey

Marital Status

- Marital status data reveal that a significant portion of the adult population is married, reflecting the traditional social structure in which marriage and family units are highly valued.

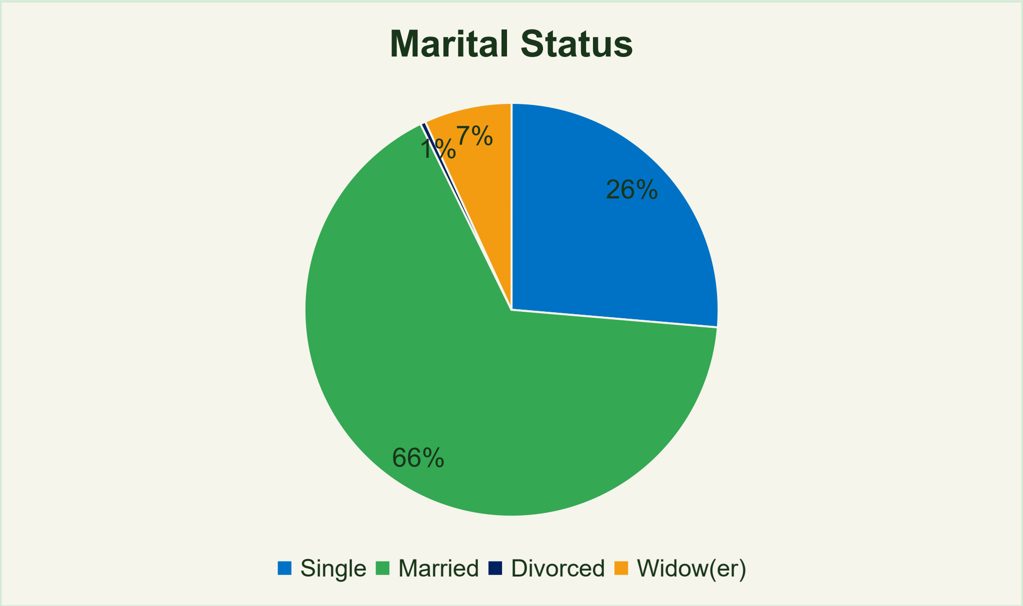


Figure 7: Pie Chart illustrating the marital status of Shao residents

Demographic Survey

Education

- Education in Shao ranges from primary to tertiary levels, with 63% having at least secondary education.
- However, 21% are illiterate, and 16% hold only primary school certificates, highlighting gapsin higher education and specialized skills.

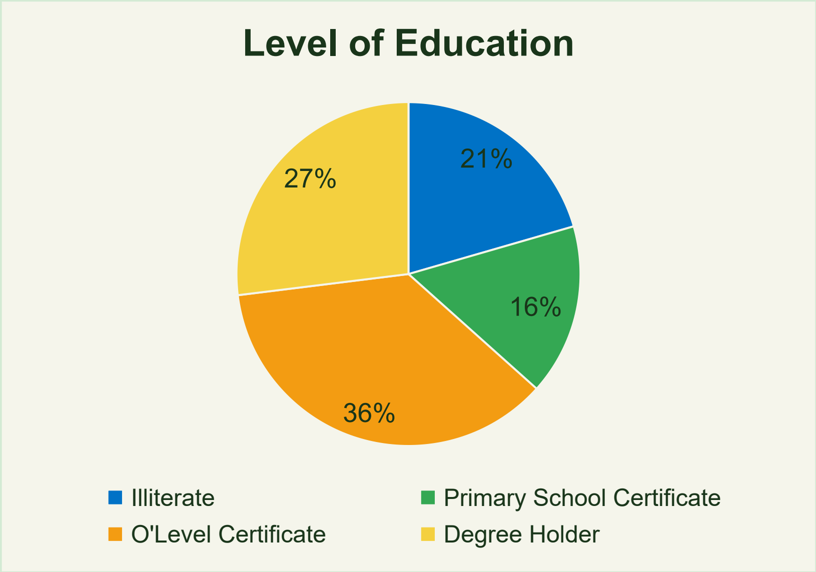


Figure 8: Pie Chart illustrating the level of education of Shao residents

Demographic Survey

Waste Disposal System

- Waste disposal in Shao is dominat242424ed by inineration (293 respondents) and open dumping (244 respondents), while landfill use is low (82 respondents).
- The absence of municipal waste collection highlights the need for better infrastructure and public awareness to reduce health and environmental risks.

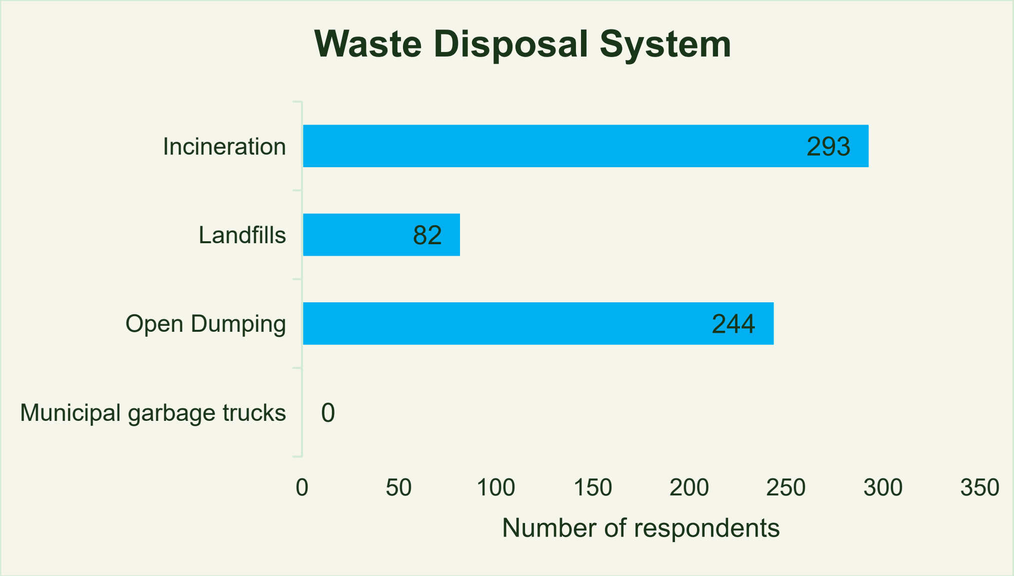


Figure 9: Bar Chart illustrating the waste disposal systems in Shao

Demographic Survey

Sewage Disposal System

- Sanitation in Shao is a major concern, with 44% practicing open defecation, posing health risks. While 37.1% use water closets, pit latrines are less common (18.9%).
- Improved infrastructure and education are needed to address this challenges.

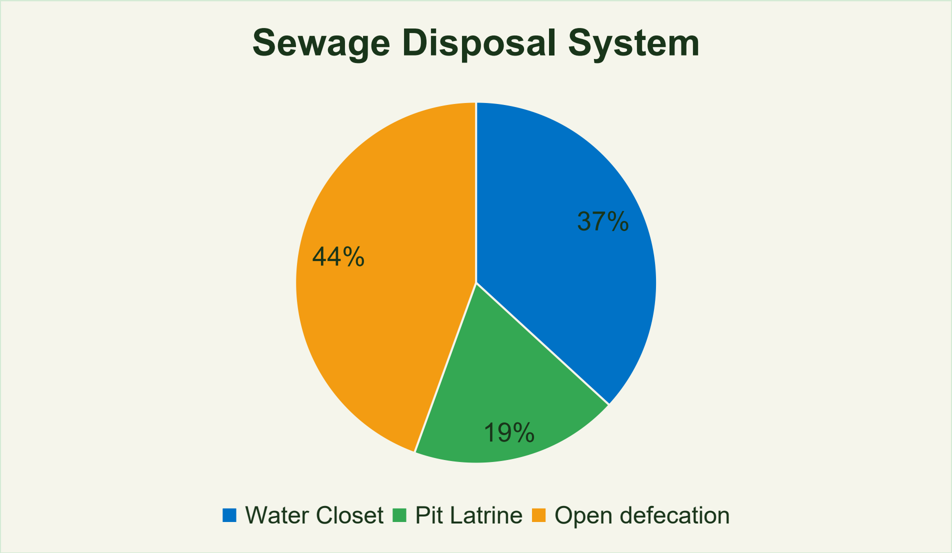


Figure 10: Pie Chart illustrating the sewage disposal system in Shao

Demographic Survey

Common Health Issues in Shao

- Health data from Shao reveals reluctance to disclose ailments, though reported conditions include malaria, ulcer, hypertension, chronic cough and arthritis.
- The PHC confirmed malaria, ulcer and chronic cough as the most treated illnessses.

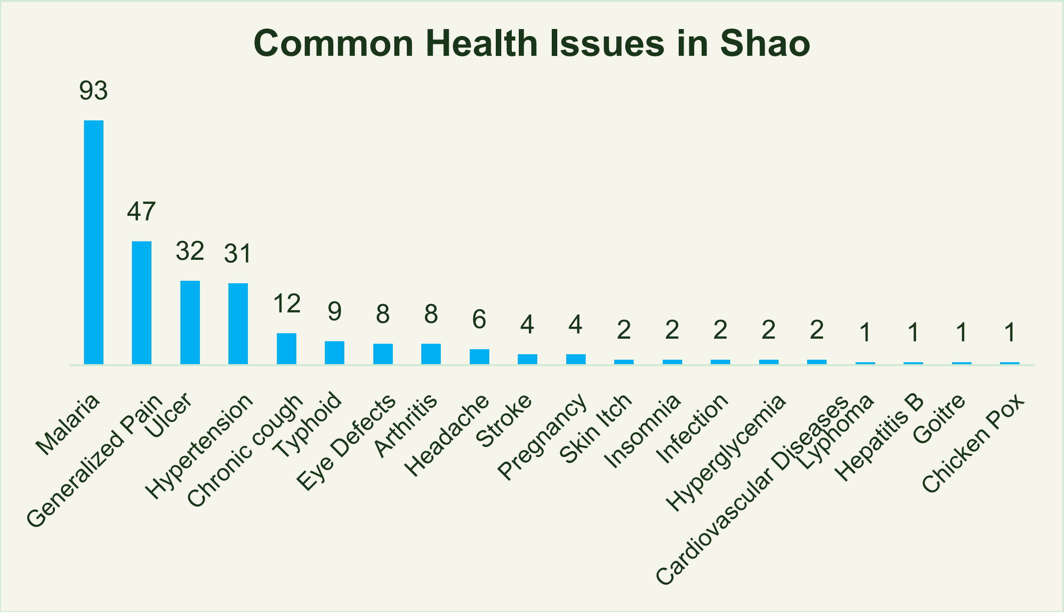
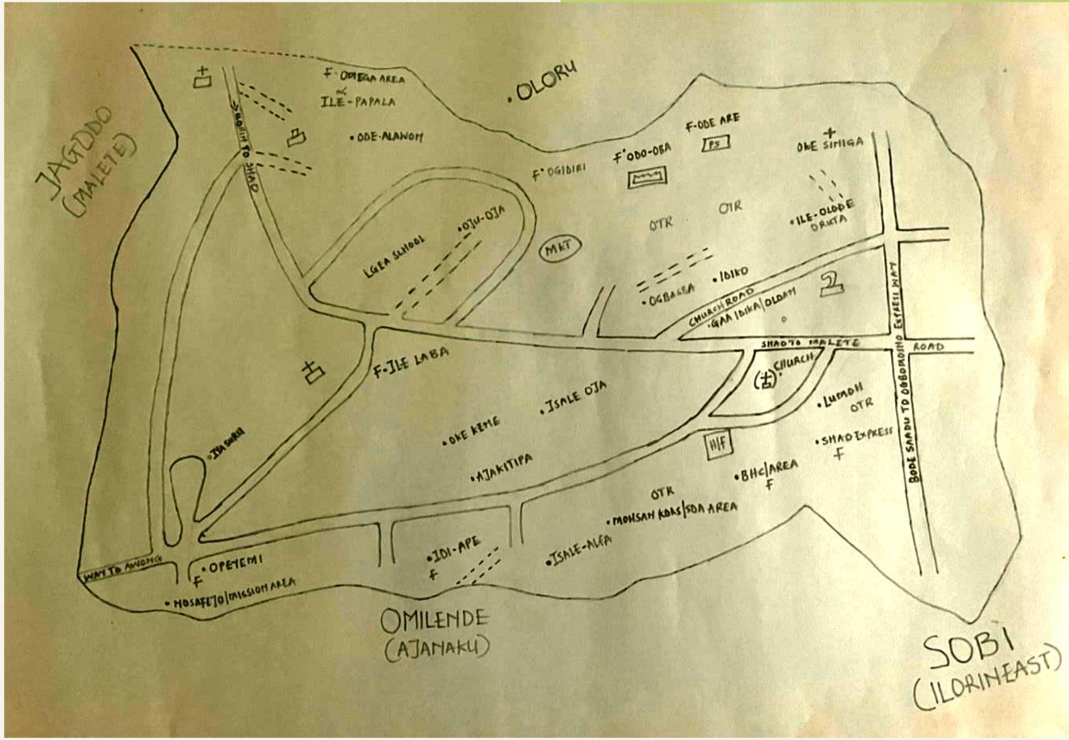


Figure 11: Bar Chart illustrating the common health issues in Shao

Map of Shao

WARDS, BORDERS AND MAJOR LANDMARKS




PRIMARY HEALTH
CARE CENTRE
SHAO
ECWA
SECONDARY SCHOOL
SHAO
Rainbow
Healthcare
PHARMACY

04

MLO 1

MLO 1: Data on Health Facilities & Services

Objective

- To Extract Data On Number and Types of Healthcare Facilities and Services Present Within The Community.

Method & Findings

- **Method**: Google Forms questionnaire (including closed and open ended questions) addressing key dimensions of healthcare.
- **Findings**: 1 PHC, 1 Private Clinic, 2 hospital based laboratories, 1 pharmacy, 14 Chemist Stores, plus unlicensed practitioners.

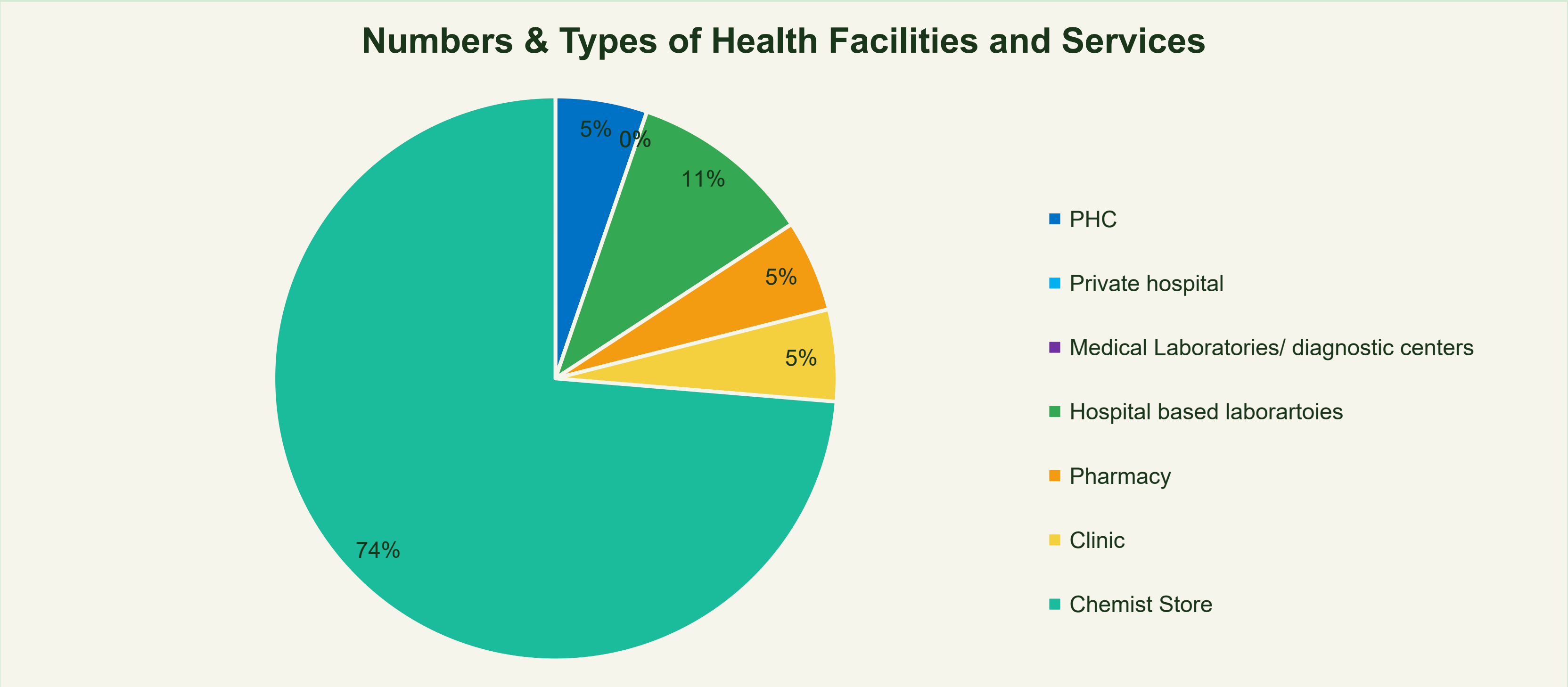


Figure 12: Pie Chart illustrating the number and types of health facilities and services in Shao community

Primary Healthcare Centre

Services offered:
- Maternal and Child Health
- Outpatient services
- Community Health Services

Limitations:
- Inadequate diagnostic services
- Workforce Shortage

The ECWA Clinic

Services offered:

- Maternal and Child Health
- Dispensation of Drugs
- Emergency Services

Limitations:

- Inadequate diagnostic services
- Workforce Shortage

The Rainbow Healthcare Pharmacy

Services offered:

- Dispensation of Drugs
- Sales of Drugs
- Prescription of medication

Limitations:

- Workforce Shortage

MLO 1: Data on Healthcare Utilization

- 229 respondents use the PHC in Shao
- 174 visit the ECWA Clinic
- 196 rely on the pharmacy and chemist stores
- 27 seek unlicensed or herbal practitioners.

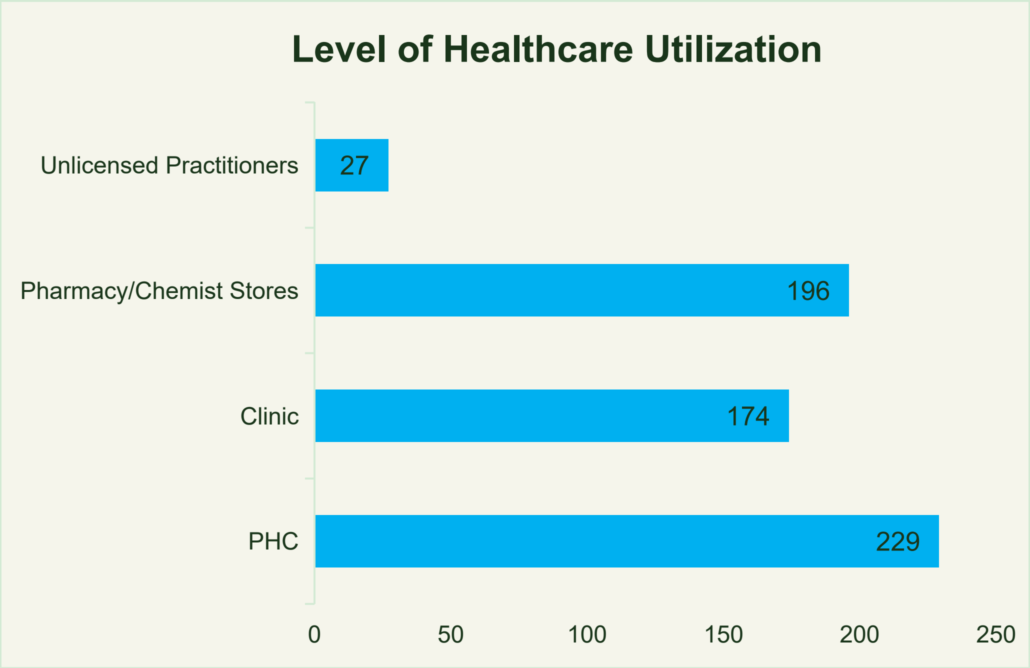


Figure 13: Bar Chart illustrating healthcare utilization in Shao

05

MLO 2

MLO 2: Challenges in Accessing Healthcare

Objective

- To determine the various Challenges Mitigating Easy Access to Satisfactory Health Services.

Method & Approach

- Community-based survey using an online questionnaire addressing the availability, accessibility, affordability, acceptability and quality of healthcare provided.

Findings

The challenges include:

- Lack of Awareness
- Lack of Infrastructure
- Lack of Accessibility and Affordability
- Cultural and Religious Beliefs
- Inadequate Laboratory Services
- Delayed Turn-around time

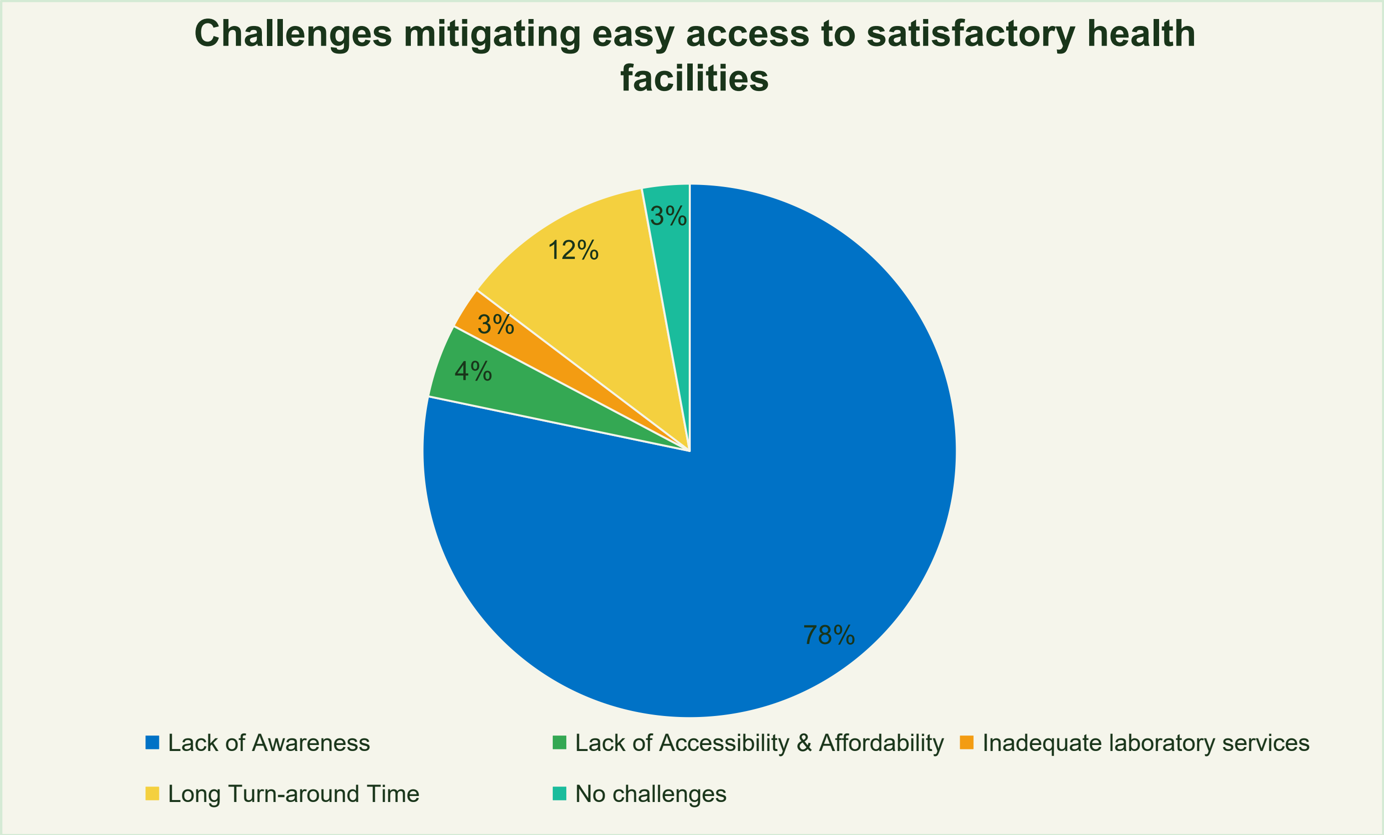


Figure 14: Pie Chart illustrating the challenges mitigating easy access to satisfactory health facilities in Shao community

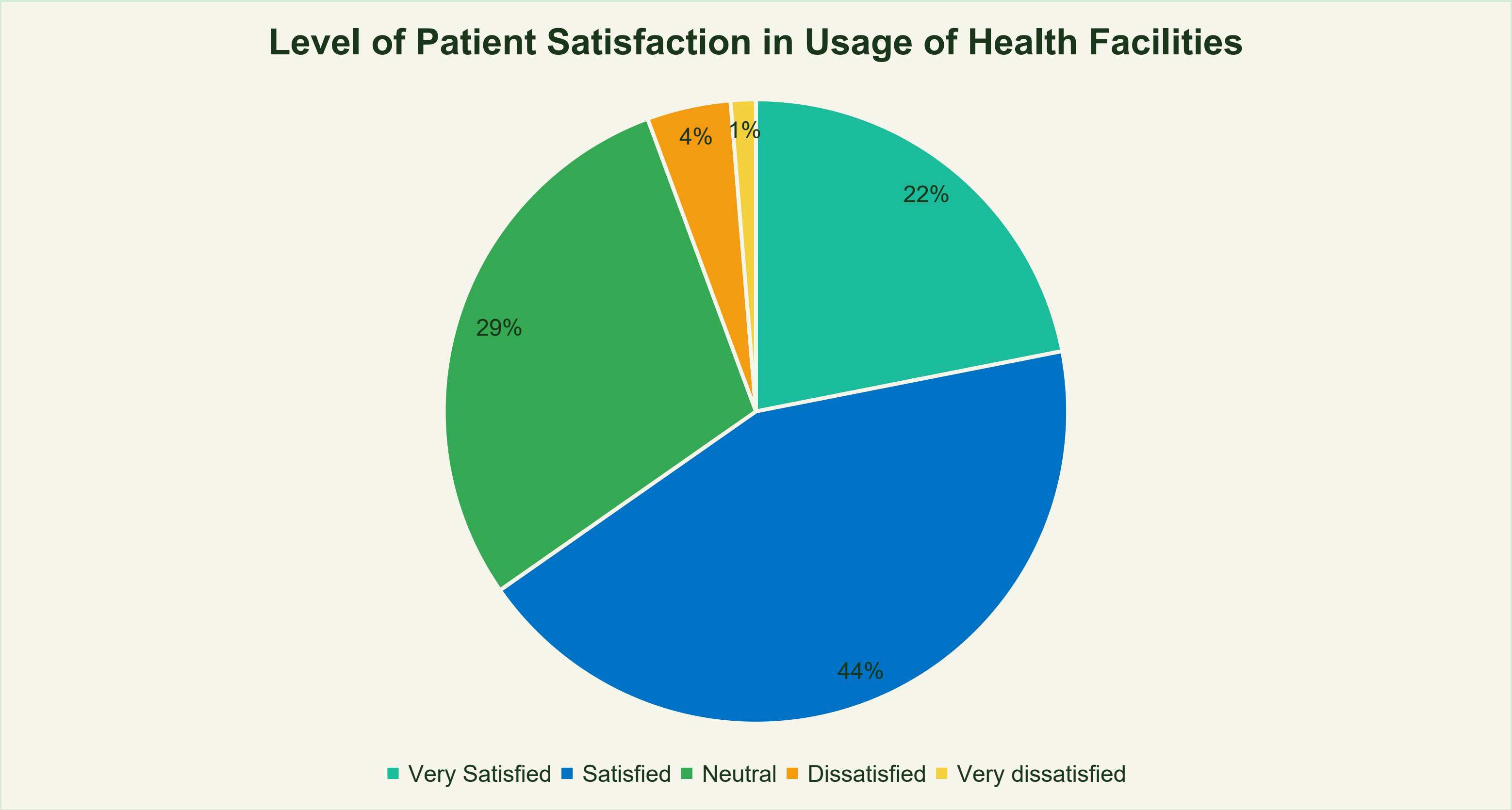


Figure 15: Pie Chart illustrating the level of patient satisfaction in their utilization of health facilities in Shao community

06

MLO 3

MLO 3: Impediments from Health Service Providers & Facilities

Objective

- To determine the impediment inherent with the Health Services Providers and Facilities making access to Satisfactory Healthcare a daunting task.

Approach

- We used a divide-and-conquer strategy, assigning each subgroup to a specific area in Shao.
- Equipped with questionnaires, data was collected from public health facilities, observed service delivery, and interviewed stakeholders.

Findings

The impediments include:

- Insufficient Personnels
- Language Barrier
- Poor Infrastructure
- Lack of Electricity

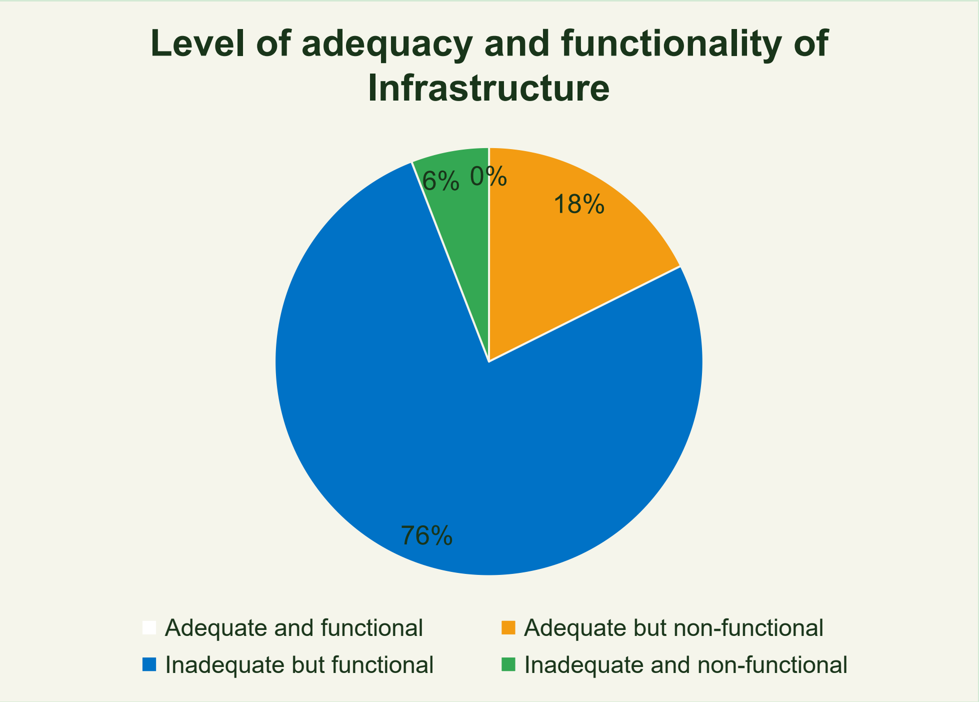


Figure 16: Pie Chart illustrating the level of adequate and functional healthcare infrastructure in Shao

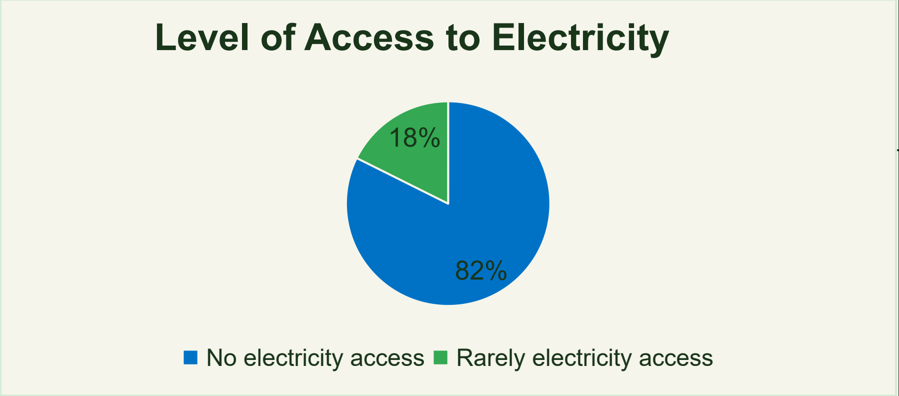


Figure 17: Pie Chart illustrating the level of electricity access by healthcare facilities in Shao

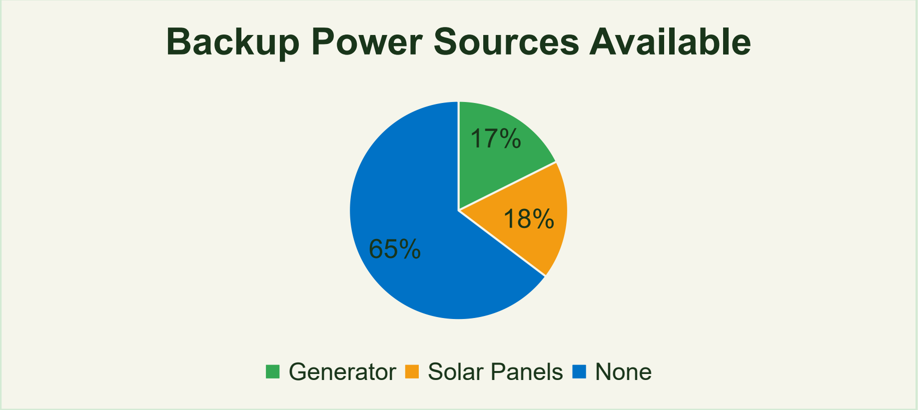
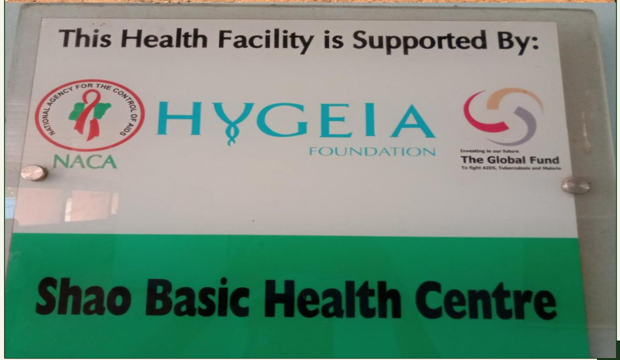


Figure 18: Pie Chart illustrating the available backup power sources by healthcare facilities in Shao


PRIMARY HEALTH
CARE CENTRE
SHAO


This Health Facility is Supported By:
NACA
HYGEIA
FOUNDATION
The Global Fund
Shao Basic Health Centre

07

MLO 4

MLO 4: Level of Government & Private Sector Involvement

Objective

- To assess the level of government and private sector involvement in health provision in the community.

Approach

- Community-based survey using an online questionnaire addressing the awareness of the community on government and private sector involvement in health provision

Findings

The government was involved in:

- Establishment of the PHC
- Construction of a public toilet

The private sector was involved in the:

- Establishment of ECWA clinic and the Rainbow healthcare pharmacy
- Provision of laboratory equipment such as microscopes by Hygeia Foundation

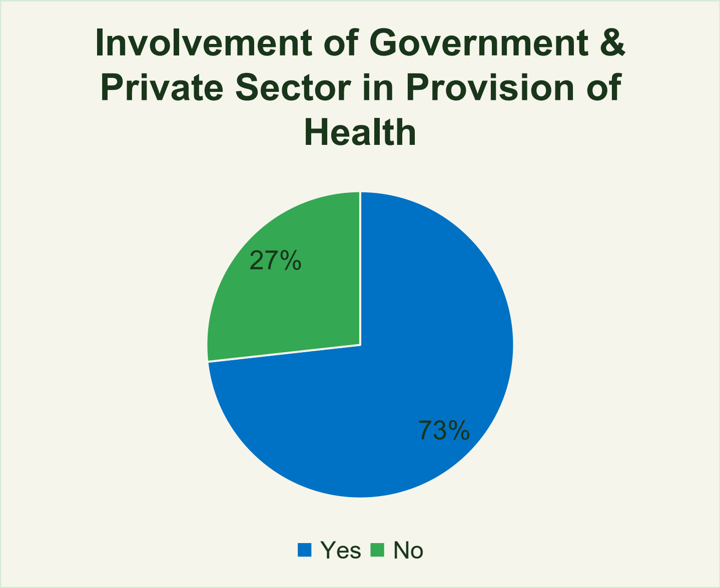


Figure 19: Pie Chart illustrating the level of government and private sector involvement in health provision based on the awareness of the residents in Shao

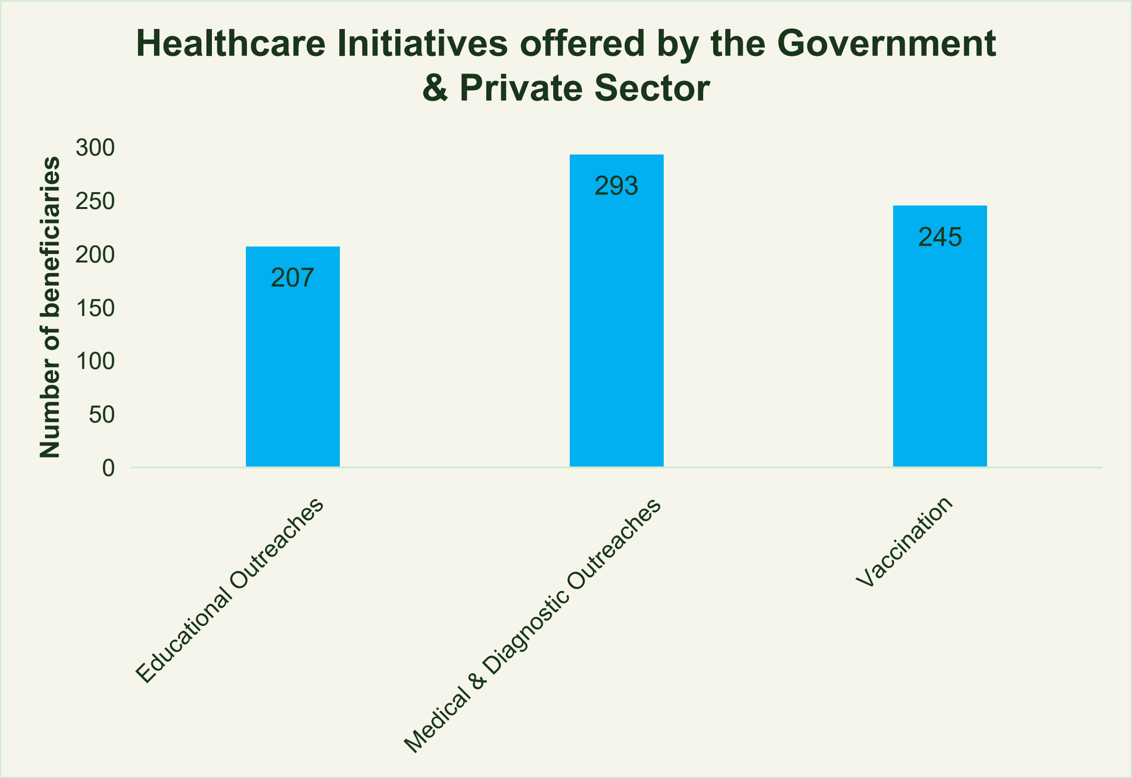


Figure 20: Bar Chart illustrating the healthcare initiatives offered by the government and private sector in Shao


TUBERCULOSIS PREVENTION
MEDICAL LABORATORY SCIENCE DEPARTMENT, UNIVERSITY OF ILORIN
IN COLLABORATION WITH WHO
COBES 2025
SHAO COMMUNITY
HEALTHY BEGINNINGS,
HOPEFUL FUTURE.
ADVOCACY FOR TUBERCULOSIS AND MALARIA
WEDNESDAY, 26TH MARCH, 2025
OHORO PALACE, SHAO, KWARA STATE.
9:00AM PROMPT
FREE HEALTH CHECKUPS
TB SCREENING
MALARIA TEST
HEPATITIS SCREENING
BLOOD PRESSURE TEST
2025
MY YEAR OF NEWNESS OF SEEKING GOD THROUGH PRAYER AND FASTING
MALARIA
SELF HYGIENE
WET HAND
USE SOAP
PALM TO PALM
FINGERS INTERLACED
BACKS OF FINGERS
CLEAN THUMBS
FINGERNAILS
RINSE HANDS
DRY HANDS

08

MLO 5

MLO 5: Sensitization & Health Promotion

Objective

- To sensitize the people and community on the importance and proactive measures to facilitate health promotion.

Approach

- Religious outreach (TB awareness at church)
- School sessions on malaria and parasitic infections
- Public rally emphasizing “No Test, No Diagnosis”
- Free diagnostic testing of the residents of Shao

MLO 5: Approach

Religious Outreach

- On March 23rd, 2025, we visited the First Baptist church in Shao and educated the congregation on tuberculosis.
- Interactive sessions used visual aids to explain symptoms, transmission, and prevention.
- The goal was to raise awareness and inspire behavioral change. Those taught were encouraged to share their knowledge.

School Outreach

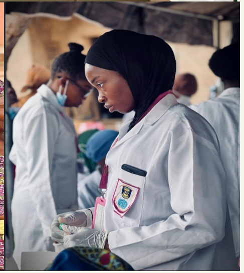
- On March 24th, 2025, a health program in Shao schools educated students on malaria, tuberculosis, and intestinal parasites. The schools reached were:
- Muslim Community Junior & Senior Secondary School
- ECWA Junior & Senior Secondary School
- 7th Day Adventist Junior & Senior Secondary School

Public Rally

- On March 25th, 2025, through a moving rally, public address system, and direct engagement, we educated the community on proper diagnosis for effective treatment and early disease detection.
- Placards reinforced key messages like “No Test, No Diagnosis,” and we invited residents to a free health outreach the next day, emphasizing proactive healthcare.

Outreach & Rally



Community Outreach

- On March 26th, 2025, we held a community health outreach at the king's palace, ensuring accessibility and high participation, with 97 individuals benefiting from free diagnostic tests.
- The event featured structured testing stations for blood pressure, BMI, malaria, tuberculosis, hepatitis B, and urinalysis, with ***Scientist Saadu I. Kolawole*** verifying the results.
- Alongside testing, we continued sensitization efforts, stressing the importance of proper diagnosis before treatment to promote early detection and better health management.

Our Findings

- The health outreach data revealed key health trends in the community. Most participants were adult females, with a high prevalence of overweight (38%) and obesity (25%), raising concerns about metabolic diseases.
- Hypertension was widespread, with 59% of participants having elevated blood pressure levels, including 10% in a hypertensive crisis.
- Urinalysis indicated potential kidney dysfunction in several cases, while infectious disease screening detected tuberculosis (1 case), malaria (4 cases), and hepatitis B (4 cases).
- These findings emphasize the need for lifestyle modifications, regular health check-ups, and continued disease surveillance.

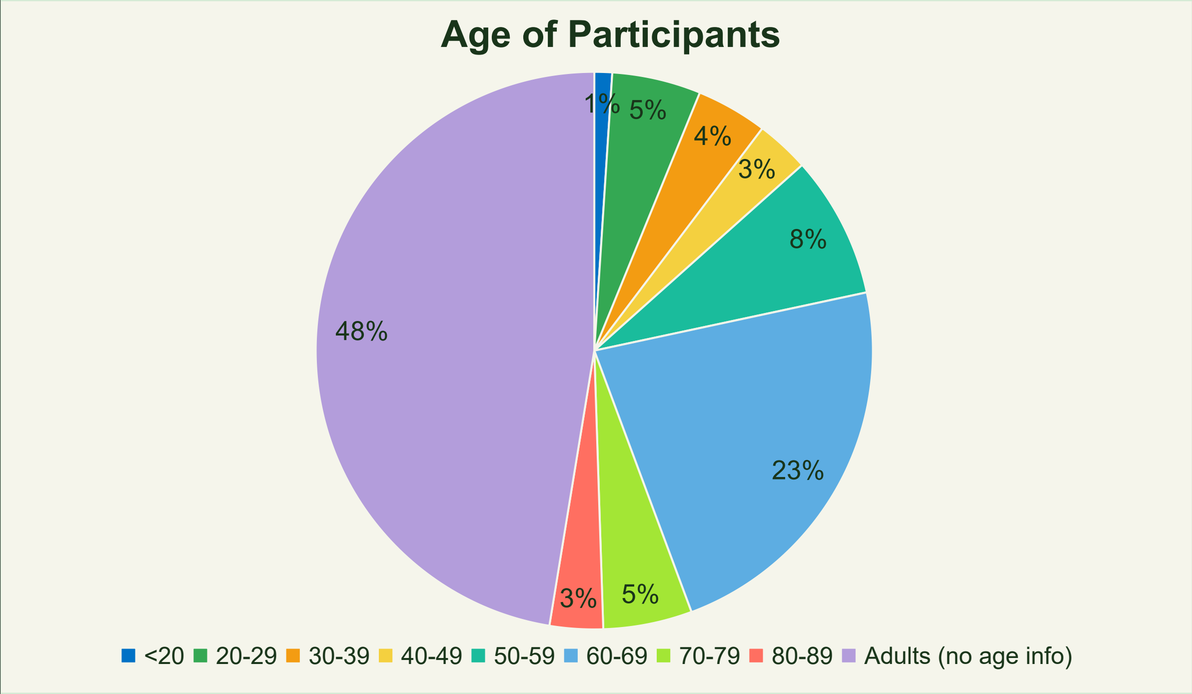


Figure 21: Pie Chart illustrating the age of the participants of the community outreach

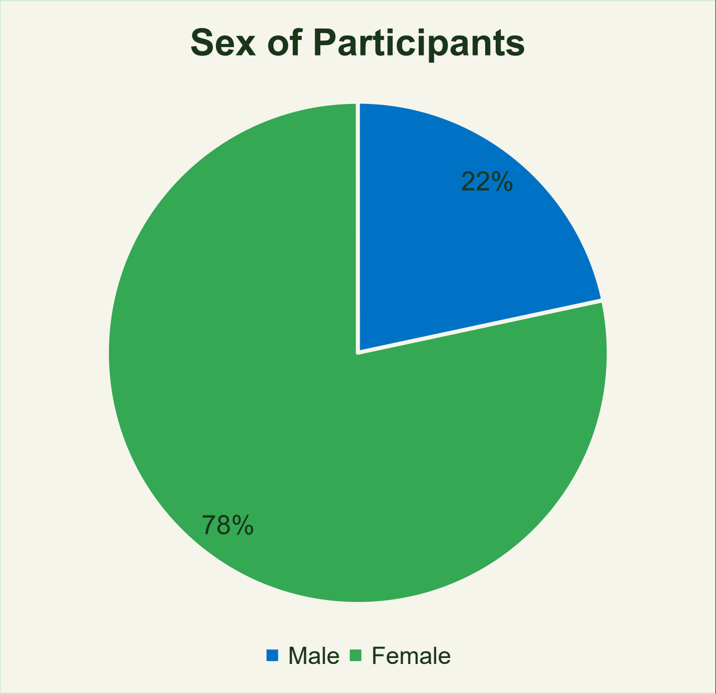


Figure 22: Pie Chart illustrating the sex of participants of the community outreach

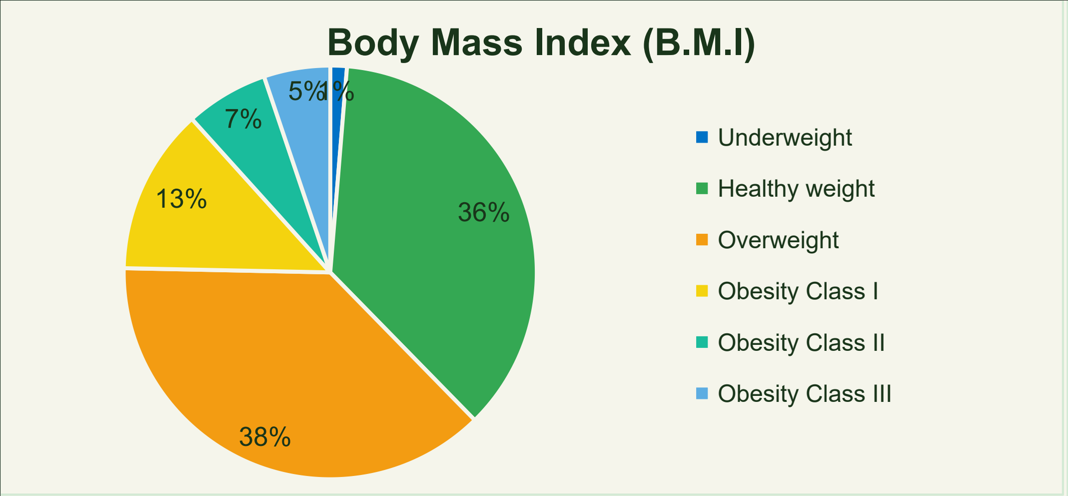


Figure 23: Pie Chart illustrating the age of the participants of the community outreach

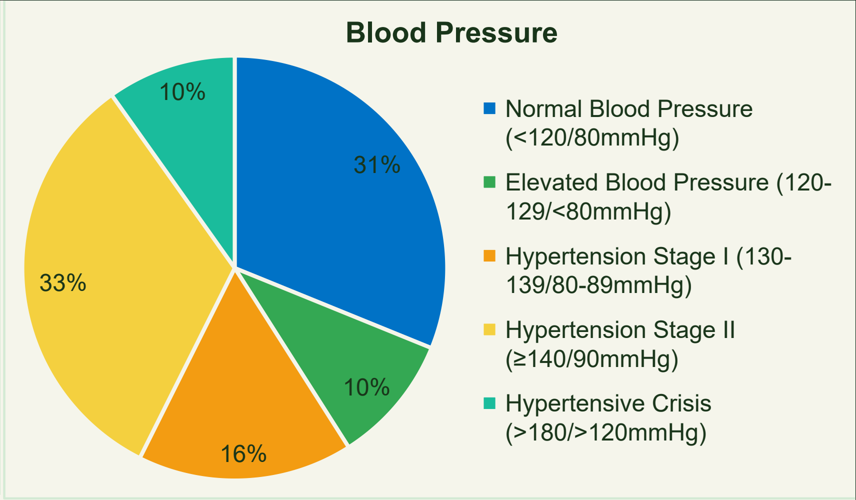


Figure 24: Pie Chart illustrating the sex of participants of the community outreach

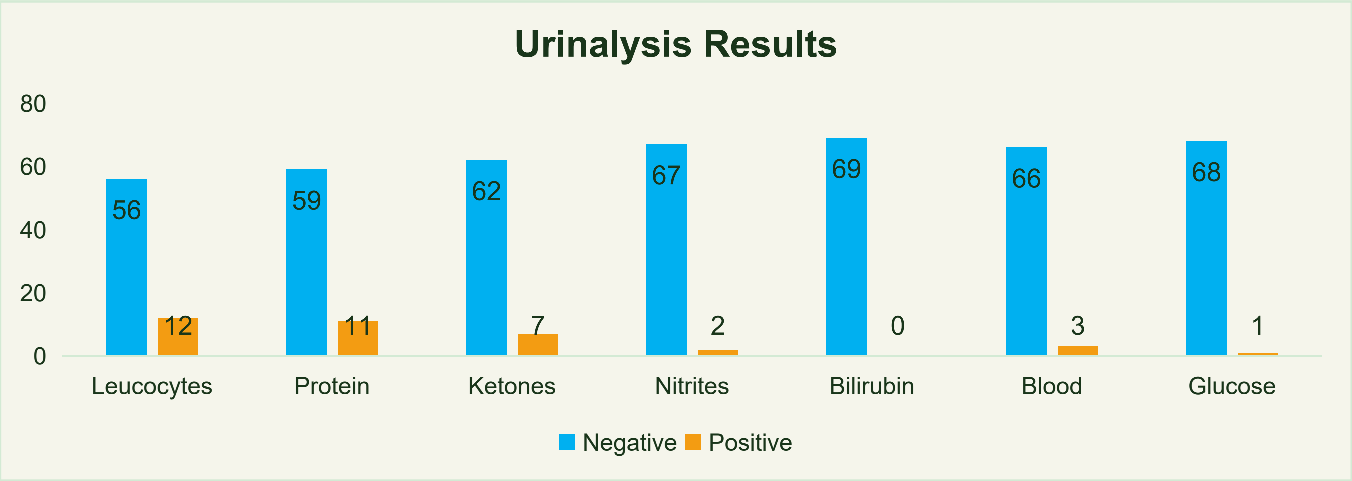


Figure 25: Bar Chart illustrating the urinalysis results gotten from the community outreach

09

MLO 6

MLO 6: Mobilizing Community Resources

Objective

- To mobilize community resources in provision of at least one project that will enhance: “it is time to build a fairer, healthier world for everyone, everywhere.”

Projects

- Incinerator construction for waste management
- Borehole renovation with generator support
- PHC lab upgrades (bench, stains, hematocrit reader)

Community Project: Incinerator & Sanitation

- Purpose: Reduce open dumping and control vector-borne diseases
- Implementation: Community mobilization and local labor

Community Project: Borehole Renovation

- Purpose: Improve water access in an electricity-challenged area
- Intervention: Cleaning, environmental renovation, generator donation
- Sustainability: Community-led maintenance plan

Community Project: Generator

Laboratory Project: PHC Laboratory Upgrades

- Objectives: Enhance diagnostic capacity
- Upgrades: Renovation of the laboratory bench, provision of Giemsa/Leishman stains, hematocrit reader
- Impact: Improved biosafety and testing accuracy

COMMUNITY BASED EXPERIENCE AND SERVICES (COBES)

COBES REPORT PRESENTATION SUBMITTED

TO COBES UNIT COLLEGE OF HEALTH SCIENCES

UNIVERSITY OF ILORIN, ILORIN, KWARA STATE, NIGERIA.

BY

500 LEVEL MEDICAL LABORATORY SCIENCES STUDENTS.

SITE: SHAO MORO LOCAL GOVERNMENT,KWARA STATE.

DATE: 17/03/2025 - 28/03/2025.

10

MLO 7

MLO 7: Reporting & Presentation

Objective

- Document findings and share outcomes.

Deliverables

- Comprehensive report and presentation to faculty and community leaders

Benefits

- Informs policy and guides future COBES initiatives

11

MLS-Specific MLO

MLS-Specific MLO: Reporting & Presentation

Objectives

To identify :

- the level of access to medical laboratory services in the community
- the gaps in the provision of effective medical laboratory services in the community
- the factors responsible for inaccessibility of medical laboratory services in the community

Approach

- Community-based survey using an online questionnaire addressing the level of access to medical laboratory services, the gaps in the provision of effective medical laboratory services, and the factors responsible for inaccessibility of medical laboratory services in the community.

Findings

- There is low level of access to medical laboratory services.
- The gaps in provision of effective medical laboratory services include: lack of infrastructure, workforce shortage, financial constraints, and limited government intervention.
- The factors responsible are: lack of awareness, cultural preference reference, lack of confidence in diagnostic services, and inadequate number of laboratories in the community.

12

Optional Project

Community Project: Sanitation

13

Optional Studies

Beske Production

- Purchase and pick soyabeans to ensure they are free of impurities.
- Soak the picked soyabeans for about 1 hour.
- After blending, sieve the shaft from the paste and extract soya milk.
- Slowly boil the soya milk while adding water and slightly fermented supernatant soya extract to aid curding.
- Use fermented supernatant as coagulant.
- Separate the curd from the boiling mixture using a cloth sieve.
- Press the curd by placing the sieve on a flat surface (wood/rock) and applying a heavy object.
- The pressed curd becomes Beske, also known as tofu or soyabean curd.
- Cut the curd into desired shapes; it can be fried, roasted, or marinated in peppered sauce for extra flavor.

Garri Production

- Harvest fresh cassava roots
- Peel the cassava roots
- Grind the cassava roots into tiny shafts using a specialized grinding machine.
- Package the grinded cassava into big sacks (rice sacks, sugar sacks).
- Press out the water from the cassava using a jack.
- Allow it to ferment for three to five days to remove cyanide.
- Filter it using a machine or manual method.
- Fry the garri thoroughly so that every grit is separated.
- Sundry the fried garri.
- Package the garri properly.

Optional Studies: Beske Production

Optional Studies: Garri Production

14 Problems Encountered & Solutions Proffered

Illiteracy

We helped the community members in filling the questionnaire, this aided information collection

Language Barrier

Members of the group who could speak some of the major languages were deployed to speak. It aided our interactions with them and made data collection easier.

Receptiveness

We respectfully introduced ourselves and explained the reason we came to their community. Our ID cards and lab coats convinced them. This made the members of the community more receptive.

Time Constrain

We divided ourselves into different groups which worked simultaneously on different aspects of the MLO to make up for the lack of time.

15 Recommendations

Improvement of Laboratory Services

Capacity Building for Health Workers

Sustainable Water and Sanitation Projects

Community Health Education

Strengthening Local Health Governance

Funding and Resource Allocation

16 Conclusion

The recommendations provided aim to create a holistic approach to health improvement in the Shao community. By addressing the identified gaps and fostering strong community involvement, we believe that sustainable health improvements can be achieved, thereby enhancing the quality of life for the residents of Shao. These measures will not only fulfill the immediate health needs but also contribute to the overall development and welfare of the community.

Thanks for listening